

CACxx variant reasonable due to relatively high availability,
FAC and DAC variants in higher demand. no real need
for 800mhz or hw-crypto for this project.

Description

STM32MP153C/F

Table 1. STM32MP153C/F features and peripheral counts

perpetually out of stock

Features		STM32MP153CADxx STM32MP153FADxx	STM32MP153CABxx STM32MP153FABxx	STM32MP153CACxx STM32MP153FACxx	STM32MP153CAAxx STM32MP153FAAxx	Miscellaneous	
		TFBGA257	LFBGA354	TFBGA361 0.5mm pin pitch	LFBGA448 0.8mm pin pitch		
Package	Body size (mm)	10x10	16x16	12x12	18x18	-	
	Pitch (mm)	0.5 ⁽¹⁾	0.8	0.5 ⁽¹⁾	0.8		
	Ball size (mm)	0.30	0.40	0.30	0.40		
	Thickness (mm)	<1.2	<1.4	<1.2	<1.4		
	Ball count	257	354	361	448		
CPU		Dual-core Cortex-A7 FPU Neon TrustZone				-	
Caches size		2 × 32-Kbyte L1 data cache					
		2 × 32-Kbyte L1 instruction cache					
		256-Kbyte level 2 unified coherent cache					
Frequency		STM32MP153C: 2 × 650 MHz STM32MP153F: 2 × 800 MHz					
MCU core		Cortex-M4 FPU				-	
	Frequency	209 MHz					
ROM		128 Kbytes (secure)				-	
Embedded SRAM		CPU system	256 Kbytes (securable)				708 Kbytes
		MCU subsystem	384 Kbytes				
		MCU retention	64 Kbytes				
		Backup	4 Kbytes (securable, tamper protected)				
SDRAM (securable)	LPDDR2/3	16-bit 533 MHz	Up to 1 Gbyte, single rank				-
		32-bit 533 MHz	-	-			
	DDR3/3L	16-bit 533 MHz	Up to 1 Gbyte, single rank				
		32-bit 533 MHz	-	-			
Backup registers		128 bytes (32x32-bit, securable, tamper protected)				-	

stm32mp153Dx:
the D variant is
for 800MHz without
hw crypto

Table 1. STM32MP153C/F features and peripheral counts (continued)

Features			STM32MP153CADxx STM32MP153FADxx	STM32MP153CABxx STM32MP153FABxx	STM32MP153CACxx STM32MP153FACxx	STM32MP153CAAx STM32MP153FAAx	Miscellaneous
			TFBGA257	LFBGA354	TFBGA361	LFBGA448	
Timers	Advanced	16 bits	2				29 timers
	General purpose	16 bits	8				
		32 bits	2				
	Basic	16 bits	2				
	Low power	16 bits	5				
	A7 timers	64 bits	2 × 4 (secure, non-secure, virtual, hypervisor)				
	M4 SysTick	24 bits	1				
	RTC/AWU		1 (securable)				
Watchdog			3 (independent, independent secure, window)				-
Communication peripherals	SPI			6 (1 securable)			-
		Having I2S		3			
	I2C (with SMB/PMB support)		6 (2 securable)			-	
	USART (smartcard, SPI, IrDA, LIN) + UART (IrDA, LIN)		4 + 4 (including 1 securable USART) some can be a boot source			Boot	
	SAI		4 (up to 8 audio channels), with I2S master/slave, PCM input, SPDIF-TX			-	
	USB	EHCI/OHCI Host	2 ports			-	
			Embedded HS PHY with BCD			-	
		OTG HS/FS (dual role port)	Yes, embedded FS or HS PHY with BCD, can be a boot source			Boot	
		Embedded PHYs		3 (2 × high-speed + 1 × full-speed)			-
	SPDIF-RX		4 inputs			-	
	FDCAN		2 (1 × TTCAN), clock calibration, 10 Kbyte shared buffer			-	
	HDMI-CEC		1			-	
	Including the following securable			1 × USART, 1 × SPI, 2 × I2C		1 × USART, 1 × SPI, 2 × I2C on securable GPIOs	
SDMMC (SD, SDIO, eMMC)			3 (8 + 8 + 4 bits), eMMC or SD can be a boot source			Boot	
QuadSPI			Yes (dual-quad), can be a boot source			Boot	

Table 1. STM32MP153C/F features and peripheral counts (continued)

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		TFBGA257	LFBGA354	TFBGA361	LFBGA448		
FMC	Parallel address/data 8/16-bit	-		4 × CS, up to 4 × 64 Mbyte		No boot	
	Parallel AD-Mux 8/16-bit	4 × CS, up to 4 × 64 Mbytes					
	NAND 8/16-bit	Yes, 1 × CS, SLC, BCH4/8, can be a boot source					Boot
Gigabit Ethernet		-		MII, RMII, GMII, RGMII with PTP and EEE		-	
10/100M Ethernet		MII, RMII with PTP and EEE					
LCD-TFT	Parallel interface	Up to 24-bit data, up to 90 MHz pixel clock (up to 1366 × 768 60 fps or up to 1920 × 1080 30 fps)					-
DMA		3 instances (1 securable), 48 physical channels in total					-
Cryptography		DES, TDES, AES-256 dual instances (secure and non-secure)					-
Hash		SHA-256, MD5, HMAC dual instances (secure and non-secure)					-
True random number generator		True-RNG, dual instances (secure and non-secure)					-
Fuses (one-time programmable)		3072 effective bits (secure, >1500 bits available for user)					-
Camera interface	Bus width	14-bit					-
GPIOs with interrupt (total count)		98		148	176	-	
	Securable GPIOs	-		8			
	Wakeup pins	4		6			
	Tamper pins (active tamper)	2 (1)		3 (1)			
DFSDM		8 input channels with 6 filters					-
Up to 16-bit synchronized ADC		2 (up to 3.6/4.5/5/6 Msps on 16/14/12/10/8-bit each)					-
	Low noise 16 bit (differential)	-		2 (1)			
	16 bit (differential)	17 (7)		20 (9)			
	ADC channels in total ⁽²⁾	17		22			
12-bit DAC		2					-
Internal ADC/DAC VREF		1.5 V, 1.8 V, 2.048 V, 2.5 V or VREF+ input					-
	VREF+ input pin	Yes					

1. With inner matrix balls having 0.65 mm pitch to allow optimized PCB routing for supplies.

2. In addition, there is also 6 internal channels for temperature, internal voltage reference, V_{DDCORE} , $V_{BAT}/4$, DAC1 or DAC2 acquisitions.